# The Geometry of SFT Perturbations: Direction, Not Magnitude, Determines Degradation

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## Abstract

Supervised fine-tuning (SFT) with LoRA perturbs pre-trained weights by a low-rank update $\Delta W$. The conventional assumption is that degradation depends on the *magnitude* of this perturbation: larger $\|\Delta W\|$ means more forgetting. Through an 84-experiment study fine-tuning Qwen2.5-3B-Instruct on math chain-of-thought data from four sources, we show this is only half the story. Within a data source, perturbation norm strongly predicts accuracy degradation ($R^2 = 0.94$ for NuminaMath, $0.86$ for NuminaMath-Hard, $0.80$ for OpenR1). But across sources, the same norm produces vastly different outcomes: NuminaMath at $\|\Delta W\| = 0.87$ yields 20% ORZ accuracy, while OpenR1 at $\|\Delta W\| = 0.79$ yields 5.4%—a 4$\times$ degradation gap at comparable perturbation magnitude. We explain this through **perturbation direction analysis**: pairwise cosine similarity between LoRA weight updates reveals that data sources occupy distinct directions in weight space (intra-source cosine $\sim$0.88, cross-source cosine $\sim$0.50), and these directions diverge as training volume increases (cosine drops from 0.75 at $N$=100 to 0.24 at $N$=10,000). We further decompose degradation into format disruption (transient, exhibiting phase-transition dynamics) and reasoning degradation (cumulative, monotonic), showing that OpenR1's direction causes catastrophic format collapse between $N$=1,500 and $N$=2,000. We validate these findings across two model scales (3B and 7B) and show that non-math SFT (code generation) produces near-orthogonal perturbation directions ($\cos \sim 0.01$), confirming the framework extends beyond math-variant data sources.

## 1 Introduction

Supervised fine-tuning (SFT) with LoRA [Hu et al., 2022] is the standard method for specializing instruction-tuned language models. A natural question is: how much degradation does SFT cause on the model's existing capabilities? The conventional answer focuses on *magnitude*—larger weight perturbations should cause more forgetting.

We present evidence that this magnitude-centric view is fundamentally incomplete. Consider two SFT experiments with comparable perturbation norms:

- **NuminaMath** $N$=10,000: $\|\Delta W\|_F = 0.87$, ORZ accuracy $= 20.0\%$
- **OpenR1** $N$=2,000: $\|\Delta W\|_F = 0.79$, ORZ accuracy $= 5.4\%$

OpenR1 achieves a 3.7$\times$ worse outcome at a *smaller* perturbation magnitude. This cannot be explained by norm alone—NuminaMath at $\|\Delta W\| = 0.60$ ($N$=2,000) maintains 23.3% accuracy, while OpenR1 at the larger $\|\Delta W\| = 0.79$ collapses to 5.4%.

We resolve this paradox by analyzing the **direction** of LoRA perturbations. Computing pairwise cosine similarity between weight update vectors across 84 checkpoints reveals three key findings:

**(F1) Data sources produce distinct perturbation directions.** Intra-source cosine similarity averages 0.88 (NuminaMath) and 0.72 (OpenR1), while cross-source similarity averages only 0.50.

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**(F2) Perturbation directions diverge with training volume.** At $N$=100, NuminaMath and OpenR1 directions are moderately aligned (cosine $= 0.75$). By $N$=10,000, they are nearly orthogonal (cosine $= 0.24$).

**(F3) Direction determines degradation rate per unit norm.** Within each source, norm strongly predicts accuracy ($R^2 > 0.80$). But the *slope* differs across sources: each unit of OpenR1 perturbation causes more damage than the equivalent NuminaMath perturbation.

**Contributions.** (1) A geometric framework for SFT degradation where perturbation direction, not just magnitude, determines which capabilities degrade. (2) Empirical evidence from 84 experiments showing source-dependent directions, $N$-dependent divergence, and source-dependent degradation rates. (3) Decomposition of degradation into format disruption (transient) and reasoning degradation (cumulative), with format disruption contributing 73% of peak loss. (4) Intervention experiments: truncating OpenR1 solutions and mixing data sources smoothly interpolates perturbation directions and outcomes. (5) Validation across two model scales (3B, 7B) and cross-domain SFT (code generation).

## 2 Related Work

**Catastrophic forgetting in continual learning.** The standard framing focuses on cross-task interference [Kirkpatrick et al., 2017, Luo et al., 2023]. EWC and experience replay are common mitigations. We show that under LoRA SFT at this scale, cross-domain forgetting is largely mitigated—the binding constraint is direction-dependent intra-domain degradation.

**Weight perturbation theory.** Random perturbation studies [Li et al., 2018] establish that neural networks tolerate perturbations below a scale-dependent threshold. Our contribution is showing that for *structured* perturbations (LoRA SFT), direction matters more than scale, with different data sources producing perturbations in distinct weight-space subspaces.

**LoRA and parameter-efficient fine-tuning.** Hu et al. [2022] showed LoRA preserves most pre-trained capabilities through low-rank updates. Biderman et al. [2024] studied LoRA for continual learning. We extend this by analyzing the geometric structure of LoRA perturbations across data sources, revealing source-dependent directional signatures.

**Data scaling and selection for SFT.** LIMA [Zhou et al., 2023] and AlpaGasus [Chen et al., 2023] study how data quantity and quality affect SFT outcomes. We add a geometric perspective: not just *how much* data, but *which direction* the resulting perturbation points determines the outcome.

## 3 Experimental Setup

### 3.1 Model and Training

We fine-tune **Qwen2.5-3B-Instruct** [Yang et al., 2024] using LoRA applied to all attention projections (Q, K, V, O) across 36 transformer layers (144 LoRA modules). Standard configuration: rank $r$=8, learning rate $5\times10^{-5}$, 1 epoch, effective batch size 16, cosine schedule, bfloat16 precision. For scale generality, we additionally fine-tune **Qwen2.5-7B-Instruct**.

### 3.2 Data Sources

All training data consists of math chain-of-thought solutions with `\boxed{}` answer formatting.

Table 1: Training data sources. All contain math CoT solutions.

| Source | Description | Mean Length | Characteristic |
|---|---|---|---|
| NuminaMath | Random sample from 860K CoT | 1,150 chars | Textbook-style, concise |
| NuminaMath-Hard | Competition problems | 2,081 chars | Detailed competition solutions |
| OpenR1 | DeepSeek R1 distillation traces | 16,469 chars | Extended reasoning, 14× longer |
| CodeAlpaca | 20K coding instructions | ∼500 chars | Code generation, non-math |

### 3.3 Evaluation

Table 2: Evaluation benchmarks and baselines (Qwen2.5-3B-Instruct).

| Benchmark | Domain | Size | Baseline |
|---|---|---|---|
| ORZ Math | Competition math | 1,024 | 28.91% |
| GSM8K | Grade-school math | 1,319 | 84.00% |
| SciKnowEval | Chemistry MCQ | 1,893 | 34.34% |
| ToolAlpaca | Tool-use | 192 | 89.22% |

We evaluate in both *strict* mode (`\boxed{}` extraction only) and *tolerant* mode (fallback to last number). The gap isolates format-dependent degradation.

### 3.4 Experiment Design

We conduct 84 experiments spanning: sample counts $N = 50$ to $10{,}000$; 4 math data sources plus CodeAlpaca; multiple LoRA configurations ($r \in \{4, 8, 16, 32\}$); 4 random seeds at key sample counts; truncation controls; data mixing at varying ratios; and 7B-scale replication.

## 4 The Norm Paradox: Same Magnitude, Different Outcomes

### 4.1 Within-Source: Norm Predicts Accurately

Within each data source, the LoRA perturbation norm is an excellent predictor of ORZ accuracy:

Table 3: Within-source norm–accuracy correlations.

| Source | $N$ Range | $\|\Delta W\|$ Range | $R^2$ | Pearson $r$ |
|---|---|---|---|---|
| NuminaMath | 50–10K | 0.06–0.87 | **0.94** | $-0.97$ |
| NuminaMath-Hard | 100–10K | 0.10–0.81 | **0.86** | $-0.92$ |
| OpenR1 | 100–10K | 0.10–1.63 | **0.80** | $-0.89$ |

### 4.2 Across Sources: Norm Fails

When sources are pooled, the picture changes dramatically. Table 4 shows key comparisons.

Table 4: Cross-source norm–accuracy comparisons. At matched norms, OpenR1 degrades $4\times$ more.

| Experiment | $N$ | $\|\Delta W\|_F$ | ORZ Acc. |
|---|---|---|---|
| NuminaMath | 1,000 | 0.48 | 23.8% |
| OpenR1 | 1,000 | 0.53 | 23.9% |
| NuminaMath | 2,000 | 0.60 | 23.3% |
| **OpenR1** | **2,000** | **0.79** | **5.4%** |
| NuminaMath | 10,000 | 0.87 | 20.0% |
| OpenR1 | 10,000 | 1.63 | 3.1% |

At $N$=1,000, both sources produce similar norms and outcomes. At $N$=2,000, OpenR1's accuracy collapses catastrophically. NuminaMath at $N$=10,000 has a *larger* norm (0.87) than OpenR1 at $N$=2,000 (0.79) yet maintains $4\times$ higher accuracy.

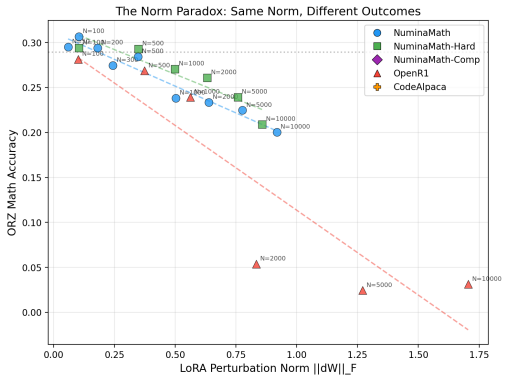
## 5 Direction Analysis: Sources Occupy Distinct Subspaces

### 5.1 Pairwise Cosine Similarity

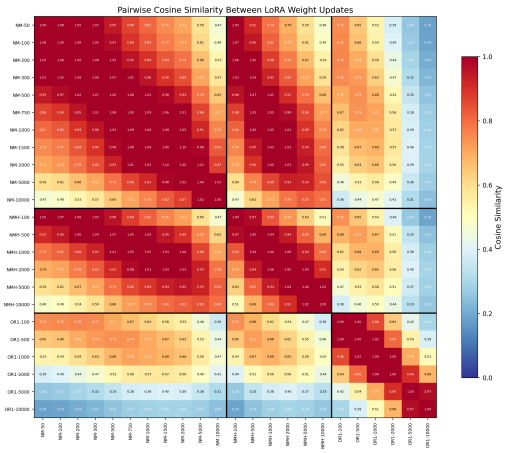
We compute cosine similarity between the full weight-update vectors ($\sim$340M dimensions) of all standard-configuration checkpoints. The results reveal strong block structure (Figure 1b).

Table 5: Average cosine similarity by source pair.

| Source Pair | Mean Cosine | Std |
|---|---|---|
| NuminaMath vs NuminaMath | 0.88 | 0.17 |
| NM-Hard vs NM-Hard | 0.86 | 0.16 |
| NuminaMath vs NM-Hard | 0.85 | 0.16 |
| OpenR1 vs OpenR1 | 0.72 | 0.23 |
| NuminaMath vs OpenR1 | **0.50** | **0.18** |
| NM-Hard vs OpenR1 | **0.48** | **0.15** |




(a) $\|\Delta W\|_F$ vs ORZ accuracy. Within-source correlation is strong, but cross-source, the same norm produces vastly different outcomes.




(b) Pairwise cosine similarity heatmap showing block structure by data source.

Figure 1: The norm paradox and directional structure. (a) Norm predicts accuracy within but not across sources. (b) LoRA weight updates cluster by data source in weight space.

### 5.2 Directions Diverge with Training Volume

Cross-source cosine similarity *decreases* as $N$ increases (Table 6). At $N$=100, all sources produce similar perturbations (cos $\geq 0.75$). By $N$=10,000, NuminaMath and OpenR1 are nearly orthogonal (cos $= 0.24$).

Table 6: Cross-source cosine similarity decreases with $N$.

| $N$ | NM vs OR1 | NM-H vs OR1 | NM vs NM-H |
|---|---|---|---|
| 100 | 0.75 | 0.75 | $\sim$1.00 |
| 500 | 0.75 | 0.72 | $\sim$1.00 |
| 1,000 | 0.71 | 0.69 | $\sim$1.00 |
| 2,000 | 0.56 | 0.56 | $\sim$1.00 |
| 5,000 | 0.36 | 0.37 | 0.93 |
| 10,000 | **0.24** | **0.26** | 0.80 |

## 6 Degradation Mechanism: Format Disruption vs. Reasoning Loss

Using strict vs. tolerant evaluation on GSM8K, we decompose total degradation into **format loss** (failing to produce `\boxed{}` answers) and **reasoning loss** (incorrect reasoning regardless of format).

### 6.1 NuminaMath: Gradual Reasoning Degradation

NuminaMath exhibits a format phase transition between $N$=500 and $N$=1,000: boxed rate drops from 99.8% to 82.0%, causing 15.5 pp of format-driven degradation. This recovers by $N$=10,000 (boxed rate 96.3%), leaving only 6.9 pp of cumulative reasoning loss.

### 6.2 OpenR1: Catastrophic Format Collapse

OpenR1 shows a qualitatively different pattern. The extended R1-style reasoning traces (16,469 chars avg) fundamentally alter the model's generation behavior.

Table 7: OpenR1 GSM8K format decomposition. The transition between $N$=1,500 and $N$=2,000 is catastrophic.

| $N$ | Strict | Tolerant | Boxed Rate | Format Loss | Reasoning Loss |
|---|---|---|---|---|---|
| 100 | 84.2% | 84.8% | 100.0% | −0.8 pp | +0.0 pp |
| 500 | 83.4% | 84.1% | 99.8% | −0.7 pp | +0.6 pp |
| 1,000 | 77.2% | 82.6% | 94.5% | +5.5 pp | +1.4 pp |
| 1,500 | 83.0% | 83.6% | 99.6% | −0.6 pp | +1.0 pp |
| **2,000** | **33.1%** | **55.3%** | **39.9%** | **+22.2 pp** | **+6.4 pp** |
| 5,000 | 33.7% | 51.9% | 37.2% | +18.3 pp | +13.8 pp |
| 10,000 | 50.0% | 60.9% | 56.4% | +10.8 pp | +12.3 pp |

The cliff is remarkably sharp: $N$=1,500 shows completely normal behavior (strict GSM8K 83.0%, boxed rate 99.6%), yet just 500 additional training samples cause catastrophic collapse (strict 33.1%, boxed rate 39.9%). Format loss alone accounts for 22.2 pp at $N$=2,000—the model can still compute correct answers (tolerant score 55.3%) but fails to wrap them in `\boxed{}`.

### 6.3 Cross-Domain Robustness

Throughout all 84 experiments, cross-domain tasks (SciKnowEval, ToolAlpaca) remain remarkably stable. Even OpenR1 at $N$=10,000 only degrades SciKnowEval by 7 pp and ToolAlpaca negligibly. LoRA perturbations are confined to a task-relevant subspace.

## 7 A Geometric Framework for SFT Degradation

We propose that SFT degradation on capability $C$ follows:

$$\text{degradation}_C(\Delta W) = g(\|\text{proj}(\Delta W, V_C)\|) \tag{1}$$

where $V_C$ is the subspace associated with capability $C$ and $g$ is monotonically increasing. The projection magnitude depends on both $\|\Delta W\|$ and the angle between $\Delta W$ and $V_C$:

$$\|\text{proj}(\Delta W, V_C)\| = \|\Delta W\| \cdot |\cos(\Delta W, V_C)| \tag{2}$$

**Predictions:** (1) Within-source linearity: if direction is constant, degradation $\approx f(\|\Delta W\|)$ alone, explaining $R^2 > 0.80$. (2) Cross-source offsets: different directions produce different projections onto $V_C$. (3) Cross-domain orthogonality: math SFT has near-zero projection onto chemistry/tool-use subspaces. (4) $N$-dependent divergence: source choice matters more at higher $N$. All four are confirmed empirically.

**Degradation efficiency.** We define $\eta = (\text{baseline} - \text{acc})/\|\Delta W\|_F$. At $N$=2,000: NuminaMath $\eta = 0.093$, NuminaMath-Hard $\eta = 0.047$, OpenR1 $\eta = 0.298$, CodeAlpaca $\eta = 0.022$. OpenR1's efficiency is 13.5× higher than CodeAlpaca despite CodeAlpaca having 1.3× larger $\|\Delta W\|$—direction dominates.

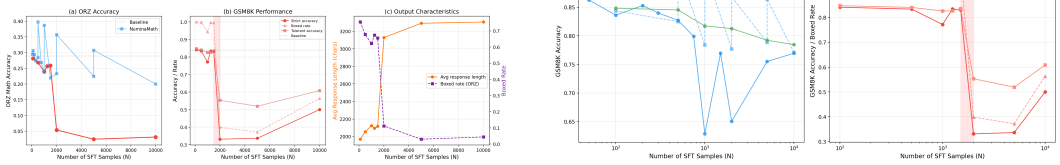
## 8 The OpenR1 Cliff: A Case Study in Directional Catastrophe

### 8.1 Characterizing the Transition

Fine-grained experiments precisely localize the transition (Figure 2a):

Table 8: The OpenR1 cliff. Three things happen simultaneously between $N$=1,500 and $N$=2,000.

| $N$ | ORZ | GSM8K (strict) | Boxed Rate | $\|\Delta W\|$ | cos(NM) |
|---|---|---|---|---|---|
| 1,000 | 23.9% | 77.2% | 94.5% | 0.53 | 0.71 |
| 1,250 | 25.7% | 83.5% | 99.7% | 0.44 | – |
| 1,500 | 25.9% | 83.0% | 99.6% | 0.49 | – |
| **2,000** | **5.4%** | **33.1%** | **39.9%** | **0.79** | **0.56** |
| 5,000 | 2.4% | 33.7% | 37.2% | 1.21 | 0.36 |




(a) The OpenR1 cliff: ORZ, GSM8K boxed rate, and response length vs $N$.

(b) Format vs reasoning decomposition for NuminaMath and OpenR1.

Figure 2: The OpenR1 cliff and format decomposition. (a) Catastrophic collapse between $N$=1,500 and $N$=2,000. (b) Format loss dominates at the cliff; reasoning loss is cumulative.

### 8.2 Seed Replication

To confirm this is not a seed artifact, we replicate with 4 random seeds. All show catastrophic collapse: ORZ $= 4.50\% \pm 0.87$ (range 3.5–5.4%). The standard deviation is small relative to the 24 pp mean drop from baseline.

### 8.3 Truncation Control

To disentangle content from style, we truncate OpenR1 solutions to NuminaMath-like length ($\sim$1,500 chars) while preserving the `\boxed{}` answer.

Table 9: Truncation control. Truncation mitigates the cliff but doesn't eliminate the gap.

| $N$ | Trunc-OR1 | Full OR1 | NuminaMath |
|---|---|---|---|
| 1,000 | 25.6% | 23.9% | 23.8% |
| 2,000 | **11.7%** | 5.4% | 23.3% |
| 5,000 | 15.2% | 2.4% | 22.5% |

Truncation substantially mitigates the collapse (11.7% vs 5.4% at $N$=2,000), but truncated OpenR1 still degrades more than NuminaMath (11.7% vs 23.3%). This shows **direction has both a content component and a style component**: style/length accounts for roughly half the gap.

### 8.4 Data Mixing

We test whether perturbation direction can be steered by mixing NuminaMath and OpenR1 at $N$=2,000 total.

ORZ accuracy interpolates smoothly (23.3% $\rightarrow$ 5.4%) while the cosine with the NM reference direction decreases monotonically (0.87 $\rightarrow$ 0.41). Format compliance shows a threshold at $\sim$75% OpenR1 fraction. This confirms direction is a continuous, manipulable quantity.

## 9 Multi-Scale Analysis: Qwen2.5-7B

We replicate core experiments at 7B scale (Figure 3a).

Table 10: Data mixing at $N$=2,000. Both accuracy and direction interpolate smoothly.

| Mix (NM/OR1) | ORZ | Boxed Rate | $\|\Delta W\|$ | $\cos(\Delta W, \text{NM-ref})$ |
|---|---|---|---|---|
| 100/0 (pure NM) | 23.3% | 82.3% | 0.60 | 0.87 |
| 75/25 | 23.2% | 82.3% | 0.65 | 0.72 |
| 50/50 | 21.3% | 82.9% | 0.71 | 0.59 |
| 25/75 | 19.9% | 70.8% | 0.75 | 0.49 |
| 0/100 (pure OR1) | 5.4% | 39.9% | 0.79 | 0.41 |

Table 11: 7B results. The OpenR1 cliff persists but shifts to higher $N$.

| | NuminaMath | | OpenR1 | |
|---|---|---|---|---|
| $N$ | ORZ | GSM8K-S | ORZ | GSM8K-S |
| 500 | 39.8% | 91.8% | 36.6% | 91.3% |
| 1,000 | 38.7% | 91.1% | 35.2% | 92.0% |
| 2,000 | 35.7% | 71.3% | 30.7% | 90.9% |
| 5,000 | 30.8% | 82.5% | **4.6%** | **56.5%** |

Three key findings: (1) NuminaMath shows smooth degradation at both scales. (2) The OpenR1 cliff exists at 7B but shifts from $N\sim$1,500–2,000 (3B) to $N\sim$2,000–5,000 (7B)—the larger model absorbs more before collapse. (3) Cross-domain benchmarks remain robust until the cliff.

## 10 Cross-Domain Directional Analysis

To test whether the framework extends beyond math variants, we fine-tune on CodeAlpaca-20k.

Code SFT directions are essentially orthogonal to all math directions (cos $\sim$0.01). CodeAlpaca has the *largest* perturbation norm ($\|\Delta W\| = 1.023$ at $N$=2,000, 1.7$\times$ NuminaMath) yet the *smallest* math degradation ($-2.2$ pp). Its degradation efficiency $\eta = 0.022$ is 13.5$\times$ lower than OpenR1's—decisively confirming that direction, not magnitude, determines degradation.

## 11 Statistical Robustness

**NuminaMath seed variations.** We run 4 seeds at $N \in \{100, 500, 2{,}000, 10{,}000\}$. Seed variance is uniformly small (0.2–1.0 pp) across all sample counts, while the degradation signal grows with $N$ (6+ pp at $N$=2,000, 9+ pp at $N$=10,000).

**OpenR1 cliff seeds.** All 4 seeds at $N$=2,000 show catastrophic collapse: ORZ $= 4.50\% \pm 0.87$ (range 3.5–5.4%), GSM8K strict $= 37.3\% \pm 3.8$, confirming this is a systematic phenomenon.

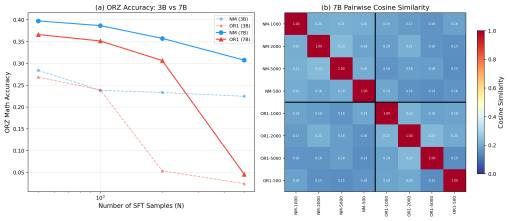
**Hyperparameter sensitivity.** Aggressive hyperparameters ($r$=16, LR= $2\times10^{-4}$, 3 epochs) amplify perturbation magnitude but preserve direction. The direction is a property of the data source, not the optimizer.

## 12 Practical Implications

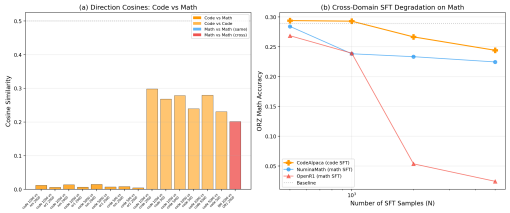
**Pre-SFT direction probes.** Train a cheap LoRA probe (small $N$) and compare its weight-update direction to known-good directions. High cosine similarity predicts safe degradation profiles.

**Data source selection.** The degradation efficiency metric $\eta$ provides actionable guidance: NuminaMath-Hard ($\eta = 0.047$) is 6.3$\times$ safer per unit norm than OpenR1 ($\eta = 0.298$).

**Format monitoring.** Separately track format compliance and reasoning accuracy. Format disruption is recoverable by training longer; reasoning degradation is not.




(a) 7B scale replication. The OpenR1 cliff shifts to $N$=5,000.




(b) CodeAlpaca directions are orthogonal to all math directions (cos $\sim$0.01).

Figure 3: Generality of directional structure. (a) Qualitative patterns preserved at 7B. (b) Cross-domain SFT confirms near-orthogonal perturbation directions.

## 13 Limitations

**Two model scales.** We test on 3B and 7B from the Qwen2.5 family. Generalization to other architectures and much larger scales remains untested.

**Capability subspace not directly observed.** We infer $V_C$ indirectly from degradation patterns. A direct characterization via activation patching or causal tracing would provide stronger evidence.

**Limited domain diversity.** We include CodeAlpaca but have not tested other domains (dialogue, summarization, multilingual).

## 14 Conclusion

We have shown that SFT-induced degradation is fundamentally a *directional* phenomenon in weight space. Within a data source, perturbation magnitude predicts degradation with $R^2 > 0.80$. Across sources, the same magnitude produces vastly different outcomes (4$\times$ degradation gap at matched norms), explained by source-dependent perturbation directions that diverge as training volume increases.

The geometric framework—degradation depends on the projection of the weight update onto capability-specific subspaces—unifies within-source norm scaling, cross-source degradation gaps, cross-domain robustness, format phase transitions, and the OpenR1 catastrophic cliff.

Our cross-domain experiments provide the strongest evidence: CodeAlpaca code SFT produces near-orthogonal perturbation directions (cos $\sim$0.01) with the largest weight update norms, yet the smallest math degradation. **Direction, not magnitude, determines which capabilities are affected.** The OpenR1 cliff reproduces across 4 seeds ($4.50\% \pm 0.87$) and at 7B scale (shifted to $N\sim$5,000), confirming it as a systematic phenomenon.

For practitioners: **not all perturbations are equal.** Choosing training data that produces perturbations aligned with the target task but orthogonal to capabilities you want to preserve is the geometric principle underlying successful SFT.

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# A Complete Experimental Results

## A.1 NuminaMath Standard Config ($r$=8, LR= $5\times10^{-5}$, 1 epoch)

| $N$ | ORZ | $\|\Delta W\|_F$ | SciKnow | TA-Real F | Valid |
|---|---|---|---|---|---|
| 50 | 29.5% | 0.056 | 34.0% | 89% | Y |
| 100 | 30.7% | 0.098 | 35.2% | 89% | Y |
| 200 | 29.4% | 0.172 | 33.4% | 89% | Y |
| 300 | 27.4% | 0.230 | 33.8% | 89% | Y |
| 500 | 28.4% | 0.330 | 35.3% | 89% | Y |
| 750 | 26.9% | 0.410 | 31.9% | 89% | Y |
| 1,000 | 23.8% | 0.478 | 32.6% | 88% | Y |
| 1,500 | 22.0% | 0.551 | 32.0% | 89% | Y |
| 2,000 | 23.3% | 0.604 | 31.6% | 89% | Y |
| 5,000 | 22.5% | 0.733 | 33.8% | 91% | Y |
| 10,000 | 20.0% | 0.869 | 31.5% | 90% | Y |

## A.2 OpenR1 Standard Config

| $N$ | ORZ | $\|\Delta W\|_F$ | SciKnow | TA-Real F | Valid |
|---|---|---|---|---|---|
| 100 | 28.1% | 0.096 | 34.6% | 89% | Y |
| 500 | 26.9% | 0.356 | 34.9% | 88% | Y |
| 1,000 | 23.9% | 0.529 | 33.7% | 86% | N |
| 1,250 | 25.7% | 0.445 | 33.4% | 89% | Y |
| 1,500 | 25.9% | 0.489 | 32.0% | 87% | Y |
| 2,000 | 5.4% | 0.789 | 35.8% | 86% | N |
| 5,000 | 2.4% | 1.211 | 30.1% | 89% | N |
| 10,000 | 3.1% | 1.628 | 27.4% | 89% | N |

## A.3 Seed Variation Details

Table 12: NuminaMath seed variations (standard config).

| $N$ | Seeds | ORZ Mean $\pm$ Std | Range |
|---|---|---|---|
| 100 | 4 | 29.6% $\pm$ 1.0 | 28.5–30.7% |
| 500 | 4 | 28.4% $\pm$ 0.2 | 28.0–28.6% |
| 2,000 | 4 | 22.6% $\pm$ 0.5 | 22.2–23.3% |
| 10,000 | 4 | 19.9% $\pm$ 0.5 | 19.0–20.5% |

Table 13: OpenR1 $N$=2,000 cliff replication across 4 seeds.

| Seed | ORZ | GSM8K Strict | SciKnow |
|---|---|---|---|
| 42 | 5.37% | 33.1% | 35.8% |
| 1 | 3.52% | 40.9% | 32.8% |
| 2 | 3.91% | 40.2% | 31.3% |
| 3 | 5.18% | 34.9% | 33.3% |
| Mean $\pm$ Std | 4.50 $\pm$ 0.87 | 37.3 $\pm$ 3.8 | 33.3 $\pm$ 1.8 |